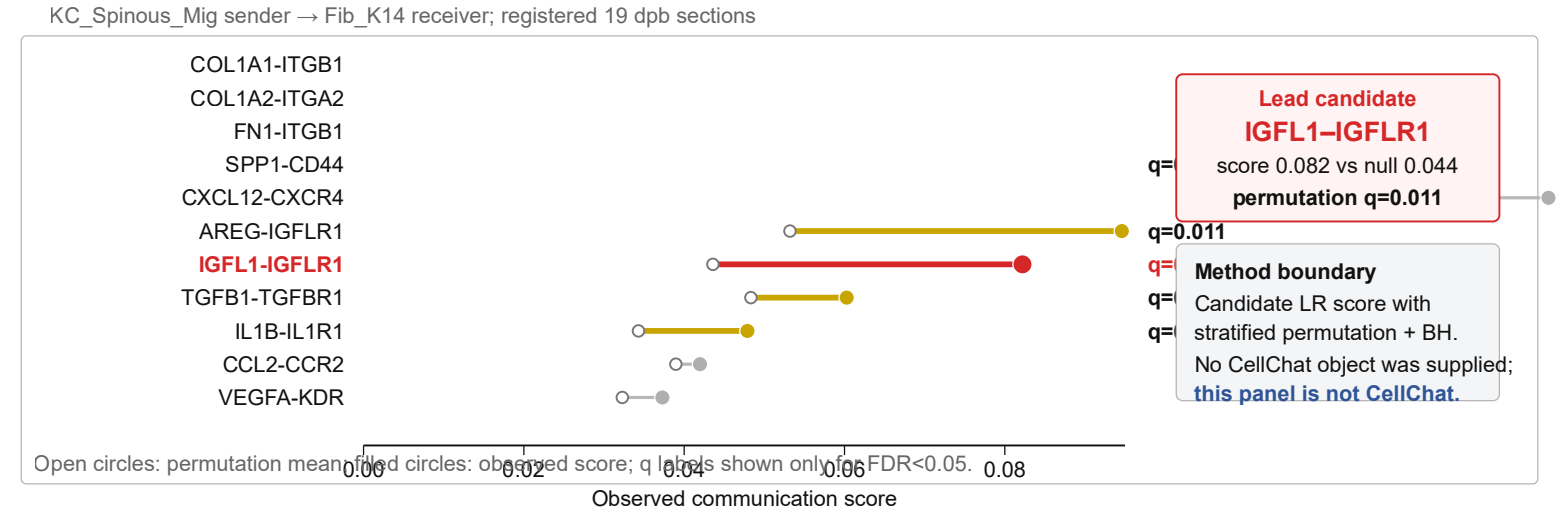
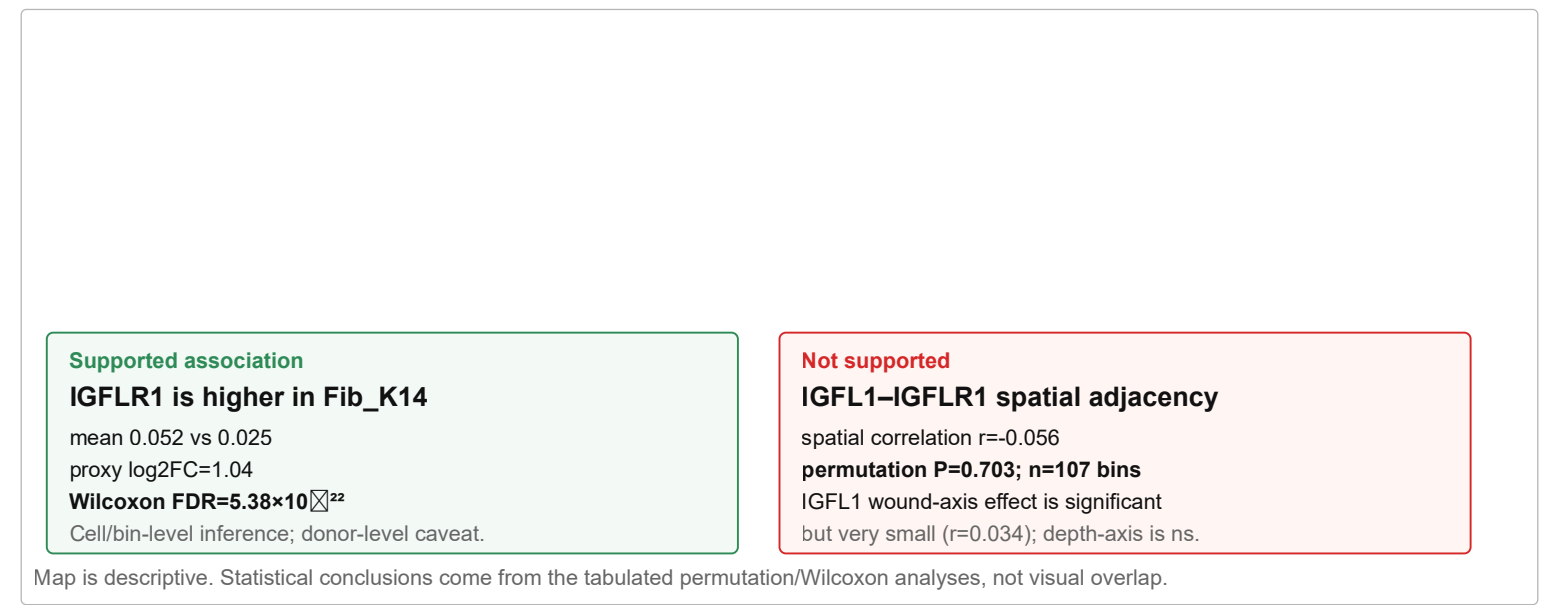


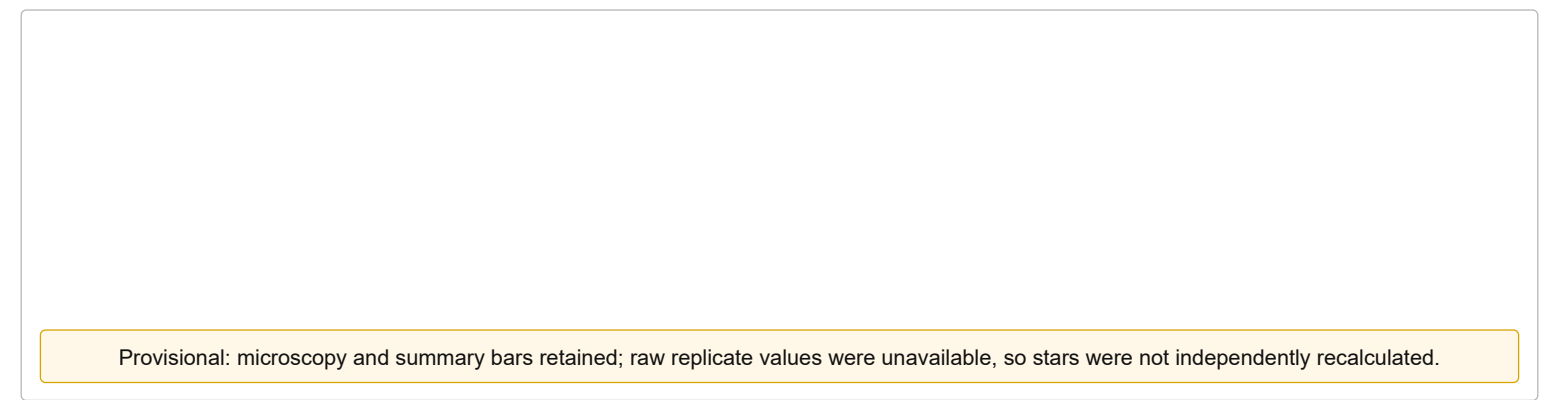
A Permutation-screened candidate ligand–receptor interactions



B Spatial distribution and independent statistical checks



C IGFL1 intervention: source-reported epithelial marker response



D Evidence-tiered IGFL1–IGFLR1–KLF4 working model

